

HARSH KUMAR

+91-8310321521 ◇ New York, NY ◇ hk3510@columbia.edu ◇ [LinkedIn](#) ◇ [GitHub](#) ◇ [Portfolio](#)

EDUCATION

M.S. in Artificial Intelligence, Columbia University Aug 2026 – Expected Dec 2027
B.Tech in Computer Science (Hons.), AI & ML, Manipal University Jaipur 2022-2026
GPA: 8.95/10 – 3× Dean’s List for Academic Excellence
Relevant Coursework: Machine Learning, Deep Learning, Reinforcement Learning, NLP, Computer Vision, Data Science, Data Structures & Algorithms, Design & Analysis of Algorithms, Object-Oriented Programming, RDBMS

SKILLS

Languages	Python, SQL, TypeScript/JavaScript, C++
AI & ML	Large Language Models (LLMs), LLM Post-Training (SFT) , Model Quantization , Inference Optimization, RAG, Model Evaluation & Benchmarking, Reinforcement Learning, NLP, Deep Learning, Multi-Agent Systems, Transformers, Computer Vision
Frameworks	PyTorch, TensorFlow, HuggingFace Transformers, LangChain, LangGraph, FastAPI, Scikit-Learn, React
Data	SQL (MySQL), NoSQL & Vector Databases (ChromaDB, Qdrant), Data Pipelines, ETL, Pandas, NumPy
Infra & Tools	AWS (Bedrock, Strands SDK), vLLM, Groq, Ollama, CUDA, Git, Linux, CI/CD, REST APIs, MCP, asyncio

EXPERIENCE

NTT Data — AI Engineering Intern (Agentic AI & ServiceNow) Dec 2025 - May 2026

- Prototyped a **production-facing agentic AI platform (LangGraph, RAG, FastAPI, ChromaDB)** for automated incident classification and evidence-grounded resolution, **deployed** against live enterprise ticket data.
- Built **AI-powered workflows and integrations** across an enterprise ITSM stack, automating multi-stage approval logic in **JavaScript** with role-based access control and full audit trails.
- Owned the feature end-to-end**, partnering directly with business and engineering stakeholders to translate ambiguous operational requirements into shipped software.

Quest Global — AI Engineering Intern May 2025 - July 2025

- Designed and **deployed a scalable retrieval-augmented data pipeline** serving multiple LLM backends (**LLaMA, Mistral**) for automated test-case generation, sustaining **80–90%** accuracy in production.
- Evaluated and benchmarked competing models and tooling** to select backends on an accuracy/latency/cost basis, iteratively refining retrieval and prompting strategy against measured results.
- Engineered an **autonomous execution agent (AWS Bedrock, Strands SDK, MCP, Playwright)** integrating third-party services into a reliable automated validation loop, cutting manual workload **60%** and release cycles **30%**.

Action Tour Guide — Python Developer Intern June 2024 - July 2024

- Built **Python ETL automation** to parse, transform, and organize high-volume customer datasets, reducing manual processing effort **40%**.

PUBLICATIONS

LoCoT: Measuring Ineffective Chain-of-Thought Reasoning Tokens in LLMs [🔗](#) *COLM Efficient Reasoning Workshop 2026*

- First-authored** work defining the **reasoning lock-in point**, a measurable property of LLM inference separating determinative tokens from wasted compute; showed up to **84%** of generated tokens are post-commitment.
- Designed a **deployable early-stopping policy retaining 88.9% accuracy at 37.1% token savings** with no oracle knowledge, against a **45–64%** oracle upper bound — a direct reduction in **inference latency and serving cost**.
- Ran a controlled truncation study over **4 models across 3 architecture families** (Qwen2.5, DeepSeek-R1, Llama-3.1) and **3 benchmarks**, spanning **4-bit NF4 quantized and bfloat16 inference**, with statistical validation ($r=0.697$ with correctness, **4–5×** stronger than prior stability metrics).

PermaFrost-Attack: Stealth Pretraining Seeding for Planting Logic Landmines [🔗](#) *arXiv Preprint, 2026*

- First-authored** a web-scale data-poisoning threat model for LLMs; designed **2 novel geometric diagnostics** surfacing adversarial influence invisible in model outputs.
- Ran the full experimental sweep, **post-training (SFT) 6 instruction-tuned LLMs (1B–14B)** on the **HH-RLHF preference corpus** and validating persistent triggerable deviations on in- and out-of-distribution prompts.

Chronocept: Instilling a Sense of Time in Machines [🔗](#) *Accepted at EACL SRW 2026*

- Co-authored the first benchmark modeling **temporal validity as a continuous probability distribution**; curated a **1,700+** sample dataset at **84–89%** inter-annotator agreement and benchmarked **RoBERTa, DeBERTa-v3, MT-DNN, SBERT** under a **Gaussian NLL** objective.

PROJECTS

AgentDesk — Agentic Triage Platform — *(LangGraph, RAG, ChromaDB, FastAPI, React/TypeScript)*

- Architected a **4-stage multi-agent pipeline** (Intake → Retrieval → Resolution → Governance) over a vector database, automating triage across a **25,000+ record** corpus.
- Designed a **dual-scoring evaluation guardrail** (embedding cosine similarity + LLM self-critique) auto-resolving above a ≥ 0.6 grounding threshold and escalating low-confidence cases with **100%** audit logging.
- Shipped a **scalable FastAPI service** against a live REST API plus a React/TypeScript trace viewer, enabling zero-code migration between mock and production environments.

Aegis — Asynchronous LLM Evaluation Framework — *(Python, asyncio, FastAPI)*



- Built a **high-throughput async evaluation harness** exercising **70+** adversarial strategies against any LLM provider (**OpenAI, Anthropic, vLLM, Groq, Ollama, HuggingFace**) behind a unified abstraction.
- Engineered a **config-driven execution engine** with **5× request concurrency** and hybrid deterministic + model-judge scoring, integrated into **CI/CD** as an automated regression gate.

MedVision — Ensemble Deep Learning Inference System — *(PyTorch, CUDA, Streamlit)*

- Trained and deployed a **3-model ensemble** (EfficientNet-B3, ResNet50, DenseNet121) on **10,000+** images at **96.22%** accuracy; optimized training via **mixed-precision (FP16)** for a **40% speedup**, serving inference at **<2s latency**.

ACHIEVEMENTS AND CERTIFICATIONS

- **Finalist**, National SAP Hackathon • **Data Structures & Algorithms** Certification, UC San Diego (Coursera) • ML Specialization, Stanford / DeepLearning.AI